Interplay of Text, Reader And Context (InTRACt) in children's digital reading comprehension: Preregistration

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Abstract

Children's reading comprehension involves the interplay of three dimensions: the text (its lexical, syntactic, and discourse characteristics), the reader (developing reading fluency, built through accumulated reading experience), and the reading context (the digital medium and the child's linguistic environment). Despite substantial laboratory-based progress in characterising each dimension in isolation, their joint influence under naturalistic, multilingual conditions is poorly understood. The present preregistration describes the InTRACt study, which addresses this gap by analysing large-scale behavioural logs from *Let's Story*, an AI-powered storytelling feature of the *Applaydu* edutainment app deployed across 19 languages. Reading time per word is the outcome. For each story, a broad range of computational text features will be extracted and reduced to principal components before model entry. Within-child reading experience — how much the child has read across prior sessions — is treated as a developing-fluency predictor, expected to speed reading as it accumulates. All confirmatory analyses use Bayesian multilevel models with random effects for child and language, so that text-feature effects are estimated across the 19 languages of the corpus. Two hypotheses are pre-registered: that syntactic and lexical complexity slow reading while discourse coherence accelerates it, and that children read progressively faster as they accumulate reading experience within the app.

1 Background

1.1 Text-Level Predictors of Reading Comprehension

A longstanding tradition in computational psycholinguistics explains reading comprehension in terms of measurable text and reader variables. This body of work identifies specific linguistic features, operating at the lexical, syntactic and discourse levels, that predict how readers engage with and understand text.

At the lexical level, word frequency is among the most reliable predictors of reading behaviour. High-frequency words are recognised faster and fixated for shorter durations than low-frequency words, a pattern replicated across dozens of eye-tracking studies and remarkably consistent across languages and orthographies: word length and frequency exert effects of strikingly similar magnitude across twelve typologically diverse alphabetic languages (Kuperman et al., 2024), evidencing the cross-linguistic regularity on which a multilingual analysis like the present one depends. In children, lexical access is less automatised, so lexical properties such as frequency and semantic diversity exert strong and still-developing effects on word reading (Hsiao & Nation, 2018). Frequency norms derived from subtitle corpora (e.g., the Subtitle Lexical Frequencies [SUBTLEX] databases), including SUBTLEX-UK for British English (van Heuven et al., 2014), have become standard tools for operationalising this variable, offering estimates that closely approximate everyday language exposure across multiple languages (Brysbaert et al., 2019). More recently, resources such as the Children and Young People's Books Lexicon (CYP-LEX) — derived from over 1,200 books read by children and young people in the United Kingdom — offer a more ecologically valid frequency baseline for studies of developing readers, capturing the vocabulary children actually encounter in their reading environment (Korochkina et al., 2024). A related lexical property, word concreteness, also shapes comprehension: concrete, highly imageable words (e.g., *apple*, *river*) are processed more quickly than abstract words (e.g., *justice*, *theory*) (Magnabosco & Hauk, 2024). This concreteness advantage, often attributed to dual-coding accounts in which concrete words activate both verbal and imagistic representations (Bernabeu, 2022; Kaup et al., 2025), is especially pronounced in younger readers and second-language learners, who rely more heavily on grounded, sensory representations to construct situation models from text — bridging gaps in background knowledge through the ease with which words evoke mental imagery (Muraki et al., 2025). Consistent with this, Pickren et al. (2022) demonstrated that text characteristics including word emotionality and concreteness predict significant unique variance in reading performance among elementary school children, above and beyond traditional readability metrics. A third lexical variable, the diversity of vocabulary within a text — operationalised through indices such as the type/token ratio (TTR) or the Measure of Textual Lexical Diversity (MTLD) (McCarthy & Jarvis, 2010) — contributes to text difficulty. Because vocabulary knowledge is itself a strong determinant of children's reading comprehension (Cain & Oakhill, 2014), the demands of lexically varied text are expected to fall most heavily on readers with weaker vocabularies, including many children.

At the syntactic level, sentence structure exerts its own demands on the reader. Measures such as mean dependency distance (the average linear distance between syntactically related words), the number of embedded clauses and subordination ratios capture aspects of structural complexity, and sensitivity to such structure is a distinct contributor to reading comprehension (Nielsen et al., 2025). Longer dependency distances, in particular, impose greater working memory demands, as readers must maintain information across intervening material before resolving a syntactic relationship

— a load that weighs heavily on children, whose syntactic processing is still developing (Delage & Frauenfelder, 2019). Converging evidence comes from Marjokorpi & van Rijt (2025), who found that explicit grammatical understanding was a robust predictor of reading comprehension in secondary-school students even after controlling for socioeconomic background, writing ability, and other literacy-related factors — underscoring that sensitivity to syntactic structure continues to underlie comprehension development well into adolescence.

Beyond words and sentences, the coherence of a text, the degree to which successive ideas are logically and semantically connected, is central to comprehension. Computational measures of coherence, including latent semantic analysis (LSA)-based sentence-to-sentence similarity, co-reference overlap and connective density, index the ease with which readers can integrate information across a passage (Crossley et al., 2019). Highly coherent texts reduce the inferential burden and are generally associated with better comprehension, particularly for readers with lower prior knowledge (Hattan & Kendeou, 2024). Complementing these deeper semantic relations are surface-level cohesion signals: explicit connectives such as *because*, *however* and *therefore* that mark the logical structure of a text. Automated tools such as Coh-Metrix (Graesser et al., 2014) now make it possible to compute many coherence and cohesion indices at once, enabling large-scale analyses of text difficulty that would be impractical to conduct by hand (Crossley, 2025).

These individual variables have also been combined into composite indices of text difficulty. Classical readability formulae (e.g., Flesch–Kincaid, Dale–Chall) aggregate surface features such as word length and sentence length into single scores, and although they are limited in linguistic sophistication, they remain widely used in educational and applied contexts (Crossley et al., 2023). More recent multi-dimensional frameworks integrate frequency, concreteness, syntactic complexity and coherence into richer characterisations of what makes a text difficult and for whom, moving beyond simple heuristics toward discourse-level representations (Crossley et al., 2023; Crossley, 2025; Goldman, 2024; Li et al., 2025).

Taken together, this body of research establishes that reading comprehension is jointly shaped by many text-level variables operating across multiple linguistic levels, alongside substantial reader differences in vocabulary and language skill (Duff & Brydon, 2020; Nation, 2019). These variables predict reading behaviour — including reading time, fixation patterns and comprehension accuracy — in both children and adults, though with developmental differences in magnitude and direction. The present study builds on this tradition by incorporating these established predictors as fixed effects in our analytic models (see Analytic Plan). We turn next to a more recent line of work that complements these feature-based approaches with neural language models.

1.2 Computational Modelling of Reading With Neural Language Models

While the text-level features reviewed above offer interpretable, theory-driven predictors of reading behaviour, a parallel line of research has begun to ask whether the representations learned by neural language models can capture the same reading phenomena — and potentially go beyond what hand-crafted features achieve. This approach treats the internal states of large language models (LLMs) as proxies for the representations that the human language system constructs during reading, and evaluates their fit to behavioural and neural data.

In adults, language models predict reading behaviour with considerable accuracy: their word-by-word predictions reproduce eye-movement patterns across different reading tasks (Hahn & Keller, 2023) and align closely with neural responses to sentences (Hosseini et al., 2024), indicating that core linguistic mechanisms — chiefly word predictability — are well captured by current models.

Extending this work to child readers presents both opportunities and challenges. Monaghan (2023) reviews how computational models originally developed for adult word reading have been adapted to simulate literacy acquisition in children, tracing the developing connections between orthography, phonology and semantics across languages. Empirical work has begun to complement these simulations with large-scale behavioural data: Sortkær et al. (2021), for instance, analysed logs from approximately 11,000 Danish schoolchildren using a reading app (BookBites) and found a moderate positive correlation ($r \approx 0.4$) between app-derived reading speed and standardised reading test scores — indicating that computational metrics extracted from naturalistic digital reading can meaningfully index developing literacy.

In sum, neural language models complement the traditional text-feature approach: where hand-crafted variables capture well-understood sources of variation, model-derived measures may capture subtler distributional and contextual regularities (Seidenberg & MacDonald, 2018). However, for child readers, the evidence base is thinner, and many of the traditional text-level variables (frequency, coherence, syntactic complexity) may still be the most informative predictors. Our study is positioned at this intersection, using both established linguistic features and a model-derived index of word predictability (surprisal) to predict children's reading behaviour in a large multilingual sample. Table 1 summarises the key studies from both traditions that inform our hypotheses.

Table 1
Representative computational-reading studies by age group

Study	Readers	Method and key findings
Hahn & Keller (2023)	Adults ($N \approx 20$, eye-tracking)	NEAT model (recurrent neural network [RNN] with attention) vs. human reading tasks. Predicted fixation patterns across tasks; task-specific differences ($d \approx 1.2$) accounted for by optimal attention allocation.

Study	Readers	Method and key findings
Hosseini et al. (2024)	Adults ($N \approx 20$, fMRI)	GPT-2 LLM (100M words) vs. brain responses. Model fully captured cortical responses; performance saturated at ~100M words.
Monaghan (2023)	Children (review)	Narrative review of computational literacy models. Adult word-reading models extend to child literacy; vocabulary growth and reading exposure interact across languages.

1.3 Reading Experience and the Growth of Reading Fluency

The reader does not stay fixed across the period of observation: children grow more fluent as they read more. The cognitive mechanism is well established — under automaticity theory, practice renders word recognition increasingly automatic, so that it demands less attention and proceeds faster, freeing resources for comprehension (LaBerge & Samuels, 1974). At the level of individual differences, the volume of reading a child accumulates is reliably associated with the growth of reading skill: a meta-analysis of print exposure from infancy to early adulthood found that how much children read predicts decoding fluency, comprehension and vocabulary, with the association strengthening across development (Mol & Bus, 2011). These relations are widely held to be reciprocal — more skilled readers read more, and more reading further builds skill — the "Matthew effect" by which early advantages compound over time (Stanovich, 1986).

The within-child direction of this relation is, however, not settled for younger readers. A longitudinal study following children from age 5 to 15 found reciprocal links between print exposure and reading skill throughout, but the effect of accumulated practice on later reading emerged only in adolescence, with skill driving amount more than the reverse during the early school years (van Bergen et al., 2021). Other longitudinal work documents substantial between-child variation in how quickly fluency develops (Khanolainen et al., 2024), and shorter-term evidence indicates that readers who read more text gain reading speed faster than those who read less (Schmidtke et al., 2024). Such effects are increasingly measurable from naturalistic behavioural logs: in primary-school children, app-logged reading time predicts reading fluency more strongly than parental self-report and retains predictive value when both are entered together (Brossette et al., 2026). This mixed but suggestive picture is what motivates testing, within the same children and at scale, whether reading time per word falls as a child accumulates reading experience across sessions (H2) — a question this naturalistic design is well placed to address, and one we frame as associational rather than developmental given the short observation window.

1.4 App-Based Data in Reading Research

Although digital reading is now widespread and much studied (Altamura et al., 2025; Furenes et al., 2021), few computational studies have used app-generated logs as a window into reading processes. The work that does exist falls broadly into two categories, which differ in their origins and scientific utility.

The first category comprises research and educational apps — platforms designed with learning objectives and, often, data collection in mind. Sortkær et al. (2021) exemplifies this approach: by analysing reading logs from BookBites, a classroom-deployed app, they demonstrated that passively recorded reading speed correlated meaningfully with standardised reading ability. Such studies establish the principle that app logs can reflect underlying cognitive processes, but they typically operate within controlled educational settings and with relatively constrained populations.

The second category, commercial 'edutainment' apps, is far less explored in the academic literature, despite its vastly larger user base. These apps combine entertainment with incidental learning goals, and their data are generated under naturalistic, unconstrained conditions. A case in point is *Let's Story*, an artificial intelligence (AI)-powered digital storytelling feature embedded within *Applaydu* — an edutainment app developed by Gameloft and Ferrero International — which enables families to co-create storybooks using Microsoft's generative AI tools across 19 languages and passively records each child's reading time (Ratner et al., 2025). Children select from a menu of story elements (plot, setting, character and theme), and the app generates a unique narrative that is then read aloud by an AI narrator in the family's chosen language. To our knowledge, no previous study has yet analysed data from Applaydu or any comparable commercial edutainment platform at scale through the lens of computational psycholinguistics. The existing literature on reading apps instead consists primarily of reviews and small-scale experiments examining specific app features, and content analyses have repeatedly found commercial educational apps to vary widely in instructional quality (Chatzopoulos et al., 2023; Furlong et al., 2025) — a caveat that applies to the Applaydu platform examined here.

This gap between the commercial availability of reading-app data and its scientific use is notable. Table 2 summarises the small number of relevant studies, along with their sample contexts and key findings.

Table 2
Selected studies on app-based reading data

Study	App type and sample	Data, focus, and key findings
Booton et al. (2023)	Educational e-books; children 3–11 yrs (11 studies)	App features: narration, augmented reality (AR), prompts, hotspots. Narration improved comprehension for pre-readers; AR supported language learning; hotspots showed no benefit.
Chatzopoulos et al. (2023)	General educational, Google Play; N = 50 apps	Rubric-based quality appraisal. Educational content was the weakest dimension; consumer ratings showed poor correspondence with expert scores.

Study	App type and sample	Data, focus, and key findings
Chuang & Jamiat (2023)	Edutainment/educational; 3–8 yr olds (50 studies)	Interactive book features. Multimedia enhancements yielded medium-to-large literacy gains; unrelated game features harmed focus.
Furlong et al. (2025)	Commercial phonics apps; $N = 309$	Content analysis of app quality. Majority lacked structured pedagogy; only ~27% met expert criteria; consumer ratings uncorrelated with expert evaluations.
Sortkær et al. (2021)	Educational (BookBites); children $N \approx 10{,}900$, grades 3–6	App reading time vs. standardised test scores. Reading speed correlated with ability ($r = .39$–$.48$); models predicted 16–18% of variance in progress.

The present project addresses this gap directly. By analysing large-scale multilingual behavioural logs from the *Let's Story* feature of Applaydu, we aim to test whether the text-level variables identified in the laboratory-based literature — frequency, lexical diversity, syntactic complexity, coherence — predict children's reading behaviour in the wild, across the 19 languages of the corpus.

These aims translate into two research questions that the confirmatory analyses address directly. The first (RQ1) concerns whether the lexical-distributional, syntactic-structural, and discourse-coherence properties of a story predict how long children spend reading each word. The second (RQ2) concerns whether children read progressively faster as they accumulate reading experience across successive sessions within the app. Each question maps onto one of the two directional hypotheses set out below, which the study design table summarises together with their statistical tests and the interpretation of possible outcomes (Table 4).

2 Methods

2.1 Participants

The study sample consists of data from the *Let's Story* feature of *Applaydu*, a commercially deployed edutainment app developed by Gameloft and Ferrero International that offers a range of activities targeting different skills. *Let's Story* is one such activity, enabling families to co-create AI-generated storybooks using Microsoft's generative AI tools across 19 languages. In the first six months following its launch in late 2024, *Let's Story* facilitated over 2.4 million story sessions and generated more than 40,000 English-language stories (Ratner et al., 2025). We will include all children (aged 7 years and older) who meet the inclusion criteria below [exact N to be determined after data extraction] (Babayiğit et al., 2022; Lervåg & Aukrust, 2010). All data are anonymised and collected passively, with no additional experimental manipulation. Children will be included if (a) parental consent was registered in-app, (b) they were aged 7 years or older as reported at registration, and (c) they completed at least 5 distinct story sessions. Sessions will be excluded if reading times fall below 2 seconds per page or exceed 5 minutes per page (indicating skipping or abandonment), if accounts are identified as duplicates via device fingerprinting, or if sessions were flagged by the app as incomplete (e.g., due to an app crash).

2.2 Data Provenance and Researcher Access

Because the *Let's Story* logs were generated by ordinary app use before this Stage 1 submission, the study is a secondary analysis of pre-existing data, and we certify the research team's level of prior access in line with the Registered Reports requirements for existing data. At the time of submission the analytic dataset has not been extracted, and no member of the team has examined any relationship between the text-feature predictors and the reading-time outcome; the only previously known quantities are aggregate platform statistics already in the public domain, such as the total numbers of story sessions and generated stories (Ratner et al., 2025). Data extraction, computation of text features, and model fitting will be carried out only after in-principle acceptance. To protect the confirmatory tests from analytic flexibility, text-feature construction and the dimension-reduction step (see Analytic Plan) will be performed without reference to the outcome variable, so that decisions about which features to retain cannot be influenced by their association with reading time. Should any unanticipated prior access to the data come to light, it will be disclosed in the Stage 2 manuscript.

2.3 Materials

The reading materials consist of short narrative passages (200–400 words each) drawn from the *Let's Story* library within the *Applaydu* app. These stories vary in vocabulary, syntax and coherence, providing natural variation in the text-level features of interest.

For each story, we will compute computational text features organised into four theoretically motivated families and extracted with tools that operate across the languages represented in the corpus. Morphosyntactic annotation — tokenisation, lemmatisation, part-of-speech tagging, and dependency parsing — will be produced with Stanza, a neural pipeline providing Universal Dependencies models for more than 60 languages (Qi et al., 2020), so that the same operational definitions apply in every language. *Lexical-distributional features* include lexical diversity (the Measure of Textual Lexical Diversity [MTLD], type/token ratio, and lexical density) and log-transformed word frequency drawn from language-specific subtitle- and corpus-based norms (e.g., the SUBTLEX family and comparable resources) (Brysbaert et al., 2019; van Heuven et al., 2014); for languages with validated norms we additionally include age of acquisition and concreteness or imageability ratings. Within this lexical-distributional family we also include, bridging the feature-based and neural approaches reviewed above, word-by-word surprisal — the negative log probability of each word given its preceding context — computed from an autoregressive language model (per-language Goldfish models; see Table 3) and averaged across the story; surprisal is among the most reliable computational predictors of reading time and bears a robust logarithmic relation to processing difficulty (Shain et al., 2024). *Syntactic-structural features* include mean dependency distance, clause count per sentence, subordination ratio, mean sentence length, and noun-phrase complexity indices (determiners and dependents per NP), all computed from the Stanza dependency graphs. *Semantic and discourse-coherence features* are led by sentence-to-sentence cosine similarity computed from language-agnostic sentence embeddings (LaBSE), which represent sentences from 109 languages in a shared space (Feng et al., 2022), complemented where available by co-reference and argument overlap and by connective density (additive, causal, temporal, and adversative connectives) derived from language-specific connective inventories. *Affective and pragmatic features* include sentiment valence, emotional arousal, and cognitive or social word-category counts, computed from validated lexica in the languages for which they exist.

Two principles govern cross-linguistic feature construction. First, each feature is defined operationally on the output of a single multilingual pipeline — Stanza for morphosyntax and LaBSE for semantic similarity — so that, for instance, mean dependency distance denotes the same computation in Finnish as in Spanish. Second, lexicon-dependent features (frequency, age of acquisition, concreteness, sentiment) exist only for languages with validated norms; a feature that is unavailable or unreliable for a language is recorded as missing and removed during pre-screening if its missingness within that language exceeds the threshold set in the Analytic Plan. All features are standardised within language before dimension reduction, and a sensitivity analysis repeats the dimension reduction separately within each language (see Sensitivity Analyses) to confirm that the pooled cross-linguistic component structure adequately represents within-language variation. This feature set will be reduced to a smaller number of principal components prior to model fitting (see Analytic Plan).

2.3.1 Operationalisation of Variables Across Languages

Table 3 specifies how each variable is operationalised and the source used to compute it. The guiding principle is to hold the measure constant across languages wherever a single validated cross-language tool or resource exists, and to fall back on language-specific validated norms — or to treat the feature as missing — only where it does not.

Table 3
Operationalisation of each text-feature variable and the source used to compute it across languages. Language-general measures (lexical diversity, all syntactic indices, coherence, and surprisal) are computed identically in every language; lexicon-dependent measures use a cross-language resource as the primary source, supplemented by native validated norms where they exist.

Variable	**Operationalisation**	**Source applied across languages**
Word frequency	Zipf-scaled log frequency, averaged over content words	`wordfreq` cross-language norms (subtitle-, Wikipedia- and web-based; 40+ languages) (Speer, 2022), validated against native SUBTLEX databases for the most frequent languages (Brysbaert et al., 2019; van Heuven et al., 2014)
Lexical diversity	MTLD (primary), type/token ratio, lexical density	MTLD index (McCarthy & Jarvis, 2010), computed on Stanza lemmas and part-of-speech tags
Age of acquisition	Mean rated AoA	Cross-language AoA ratings (Łuniewska et al., 2016) and native databases where available (e.g., English (Kuperman et al., 2012)); treated as missing where coverage is insufficient

Variable	Operationalisation	Source applied across languages
Concreteness / imageability	Mean rated concreteness	Validated human norms where published (e.g., English (Brysbaert, Warriner, et al., 2014)); treated as missing otherwise
Word predictability	Mean per-word surprisal (negative log probability)	Comparably trained per-language autoregressive models covering 350 languages (Goldfish) (Chang et al., 2024); logarithmic surprisal–reading-time link (Shain et al., 2024)
Dependency distance, clause count, subordination ratio, sentence length, NP complexity	Counts and ratios derived from dependency parses	Stanza neural pipeline (Qi et al., 2020) under the Universal Dependencies v2 scheme (Marneffe et al., 2021)
Sentence-to-sentence coherence	Mean cosine similarity of adjacent sentences	Language-agnostic sentence embeddings (LaBSE) (Feng et al., 2022)
Co-reference and argument overlap	Lemma and named-entity overlap across sentences	Stanza lemmatisation and entity recognition (Qi et al., 2020)
Connective density	Counts of additive, causal, temporal and adversative connectives	Language-specific connective inventories (cf. (Crossley et al., 2019))
Valence and arousal	Mean rated valence and arousal	NRC VAD lexicon and its multilingual translations (Mohammad, 2018), validated against native affective norms where published (e.g., English (Warriner et al., 2013))
Cognitive and social word categories	Proportional word-category counts	LIWC dictionaries — LIWC-22 for English (Boyd et al., 2022), with published translations for other languages

These resources cover the languages represented in the *Let's Story* corpus (see Participants). The language-general measures — lexical diversity, all syntactic indices, sentence-to-sentence coherence, and surprisal — are computed identically in every language by the same tools and therefore constitute a single measure applied across languages. Among the lexicon-dependent measures, word frequency (`wordfreq`) and valence and arousal (the NRC VAD lexicon and its translations) are likewise available for every language in the sample from one cross-language resource, with native validated norms — the SUBTLEX frequency databases and language-specific affective norms — used for the five most frequent languages as a convergent-validity check. Age of acquisition, concreteness, and the LIWC word-category counts have validated coverage for only a subset of languages; where a norm is unavailable or unreliable for a language, the corresponding feature is recorded as missing and removed by the pre-screening rule (see Analytic Plan), and the per-language PCA sensitivity analysis confirms that such omissions do not distort the component structure. Where robust tokenisation or dependency parsing is unavailable for a language (for example, a language without reliable automatic word segmentation), the affected features are flagged and, if necessary, that language is excluded from the relevant analyses.

The *Let's Story* feature of Applaydu records several behavioural variables for each reading session: reading time (in seconds) per page and per story, interaction events (clicks, page flips, hint usage), and session metadata (language, an age flag, date/time). The platform does not record whether read-aloud narration was enabled, nor a comprehension measure that is comparable across the languages in the corpus, nor the auxiliary child- and family-level measures collected by other *Applaydu* activities (for example, a vocabulary score, a self-reported difficulty rating, a joint-engagement measure from shared play, or caregiver age). Child age is available only as a single "7 and older" band, with no finer resolution, so it serves solely as an inclusion criterion and cannot enter the models as a predictor or moderator. The analysis therefore rests on reading time and the session metadata, with reading time per word as the sole outcome.

2.4 Procedure

Reading occurs under naturalistic, unconstrained conditions ("in the wild"). We will restrict the analysis to first-time readings of each story (excluding repeated readings) and require a minimum of 5 distinct stories per child. Data extraction will preserve the chronological order of sessions but exclude all personally identifying information. To reduce the influence of extreme outliers, reading time values will be winsorised at the 1st and 99th percentiles within each language group.

2.5 Measures

The outcome is reading time per word, computed as total story reading time divided by word count. Because the platform records no comprehension measure that is comparable across the languages of the corpus, reading time per word is the sole outcome and no secondary comprehension outcome is analysed.

The predictor variables fall into two categories. Text features comprise the four families of computational indices described in Materials; these will be reduced to principal component scores before entering any model (see Dimension Reduction via Principal Component Analysis). The reader feature is within-child reading experience, operationalised primarily as the lagged, log-transformed cumulative number of words the child has read in prior sessions (with session order, the sequence position, retained as a robustness coding).

The resulting dataset has a multilevel structure. At Level 1, observations correspond to individual story-reading instances (one child reading one story). At Level 2, observations are nested within children. At Level 3, children are nested within languages. Text-feature principal component (PC) scores will be grand-mean centred after computation, and reading experience (cumulative words read) will be centred within child (person-mean centred), so that its effect is estimated purely within child and is separated from between-child differences in total app usage.

Two properties of the data condition the inferences these measures can support. First, reading time per word indexes reading effort only to the extent that the child sets the pace and engages with the text; we therefore verify that recorded reading time covaries with text length as a precondition for analysis (see Outcome-Neutral Quality Checks and Positive Controls), since without that relationship recorded times could not be read as an index of engagement with the text. Second, because the data are observational, neither the child's language nor the amount the child has read is experimentally assigned, and accumulated reading experience is further confounded with selective retention, in that children who remain in the app read more. We therefore frame the within-child experience effect as associational rather than causal — estimating it within child and net of the text-feature components, and accompanying it with a completer-only sensitivity analysis (see Sensitivity Analyses) — and report the selection limitation explicitly (Kraemer et al., 2000). These constraints define the causal scope of the confirmatory tests and are revisited when each hypothesis is interpreted.

Figure 1 provides a schematic overview of the study's conceptual framework, relating the text features and within-child reading experience to the reading-time outcome and the two preregistered hypotheses.

2.6 Analytic Plan

All confirmatory analyses will use Bayesian hierarchical models fitted with the `brms` package (Bürkner, 2017), which implements full Bayesian inference via Hamiltonian Monte Carlo (No-U-Turn Sampler). We adopt this framework in preference to null-hypothesis significance testing because our scientific questions concern the *magnitude* and *direction* of effects rather than their mere existence, the expected sample size confers effectively unlimited frequentist power that renders p-value thresholds uninformative, and a single unified framework avoids the family-wise error inflation that arises from conditional multi-stage testing. Prior distributions are specified to be

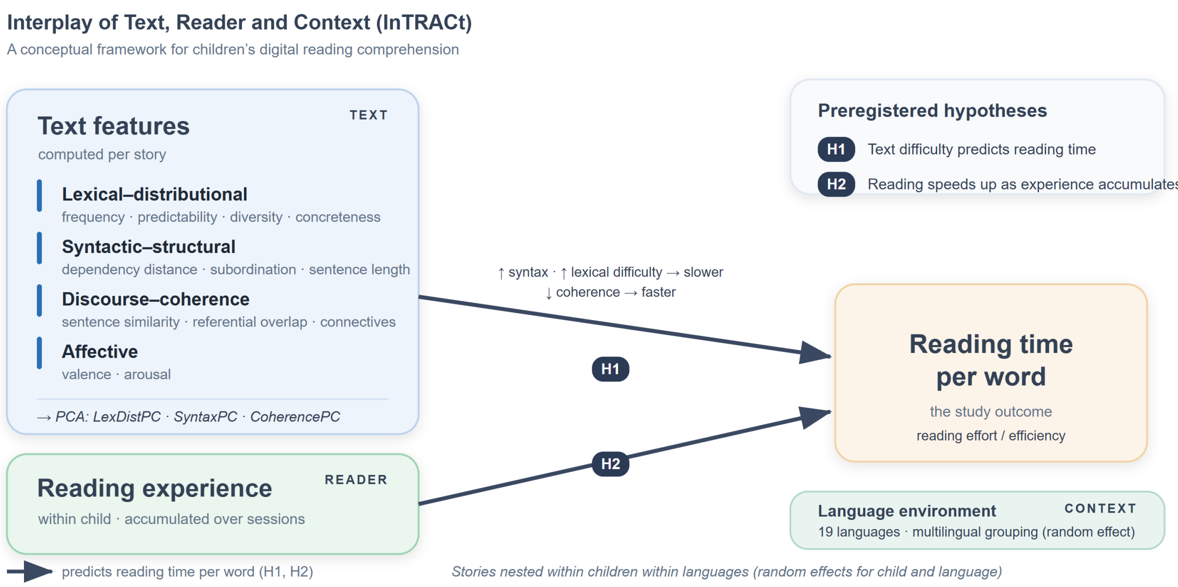

Figure 1: Conceptual framework of the InTRACt study. For each story, computational text features in four families — lexical-distributional (word frequency, predictability, lexical diversity, and concreteness), syntactic-structural (dependency distance, subordination, and sentence length), discourse-coherence (sentence-to-sentence similarity, referential overlap, and connectives), and affective (valence and arousal) — are reduced to principal components (LexDistPC, SyntaxPC, CoherencePC). These text-feature components predict the outcome of reading time per word (H1). Accumulated within-child reading experience — how much the child has read across prior sessions — is a second predictor of reading time (H2), expected to speed up reading as a child reads more. Whether reading experience additionally moderates the text-feature effects is examined as an exploratory question. Stories are nested within children within languages, with random effects for child and language, so that the analysis spans the 19 languages of the corpus.

genuinely informative; their motivation and the prior placed on each parameter class are set out under Prior Specification below.

For each pre-registered hypothesis we will report, unconditionally and regardless of outcome, the posterior mean and 95% highest-density interval (HDI) for the relevant parameter; the posterior probability that the parameter falls in the predicted direction, P(β < 0) or P(β > 0) as specified per hypothesis; and the proportion of the posterior that lies within a region of practical equivalence (ROPE) (Kruschke, 2018) defined as $|\beta| < 0.05$ on the standardised scale — the threshold below which we consider an effect too small to be educationally meaningful. If more than 95% of the posterior lies within the ROPE we conclude practical equivalence; if less than 5% lies within the ROPE we conclude the effect is practically meaningful. Both confirmatory hypotheses are evaluated under this framework simultaneously and unconditionally, with no first-stage filter. Four chains of 4,000 iterations each (2,000 warm-up) will be run, with $\hat{R}$ < 1.01 and effective sample size > 1,000 per parameter as convergence diagnostics.

2.6.1 Prior Specification

We specify genuinely informative, regularising priors. This departs from the diffuse or default priors common in applied work; the default `brms` prior on population-level coefficients, for instance, is an improper flat prior that imposes no regularisation (Bürkner, 2017). This choice is motivated on two grounds. First, priors that concentrate mass on plausible parameter values regularise the posterior geometry and thereby improve Hamiltonian Monte Carlo sampling — reducing divergent transitions, raising effective sample sizes, and shortening warm-up and total sampling time — an instance of the principle that computational difficulties usually signal an under-constrained model rather than a purely numerical obstacle (Betancourt, 2017; Gelman et al., 2020). For a maximal random-effects structure over many-levelled and partly sparse Child and Language groupings, this regularisation is also what makes a complex model estimable within a feasible time frame, and more informative priors are known to speed convergence in exactly such hierarchical, reading-time models (McElreath, 2020; Schad et al., 2021). Second, although the very large sample means the likelihood will dominate the posterior for the main effects (Lemoine, 2019), the priors still do real work in the sparsely populated parts of the design — small language strata and the by-child random slopes — where they guard against implausibly large estimates.

Because the predictors are standardised and reading time is log-transformed, the fixed-effect coefficients are interpretable on a per-standard-deviation scale, on which reading-research effects are modest: word frequency and predictability enter reading time approximately logarithmically and with small per-unit slopes (Shain et al., 2024), and app-based reading-speed effects are moderate ($r \approx .39$–$.48$) (Sortkær et al., 2021). We therefore place Normal(0, 0.2) priors on the population-level coefficients. Normal rather than Student's t priors are used for these coefficients because their log-concave tails additionally aid sampling (Gelman et al., 2020). Because both hypotheses are evaluated confirmatorily and unconditionally, every prior is centred at zero and symmetric: it regularises through its scale alone and does not encode the predicted direction, so the data remain free to disconfirm each hypothesis (Schad et al., 2021). Group-level standard deviations (for Child, Language, and the random slopes) receive half-Normal(0, 1) priors, as does the residual standard deviation; the random-effect correlation matrices receive LKJ($\eta = 2$) priors, which place the mode at the identity matrix and gently discourage the near-perfect random-slope correlations that drive divergences in maximal models (Bürkner, 2017; Lewandowski et al., 2009).

All priors will be examined with prior predictive simulations before any model is fitted to the data, and their influence will be quantified in a pre-registered prior-sensitivity analysis that compares them against the weakly informative alternatives (see Sensitivity Analyses), in line with recommended Bayesian-workflow practice (Schad et al., 2021; van de Schoot et al., 2021).

2.6.2 Sampling Plan and Inferential Sensitivity

Because the study analyses pre-existing observational data, the sampling plan is defined by the inclusion and exclusion criteria (see Participants) rather than by recruitment, and the analytic sample — first-time story readings nested within children, who are nested within languages — is expected to be very large: in its first six months the platform recorded over 2.4 million story sessions and more than 40,000 generated English-language stories (Ratner et al., 2025). The exact analysable N is fixed by the pre-registered filters and is unknown until extraction. At this scale the binding constraint on inference is the precision of estimation and the *practical* magnitude of effects, rather than statistical power: even negligible effects attain conventional significance when N is in the millions (Gelman & Carlin, 2014). We therefore replace a conventional power analysis with a precision-based Bayesian design analysis organised around the region of practical equivalence (ROPE) and the highest-density-interval decision rule (Kruschke, 2018).

Prior to fitting any model to the real data, we will conduct a simulation-based design analysis. We will generate synthetic datasets that match the planned multilevel structure (children within languages), realistic per-child and per-language session counts, and a range of plausible effect sizes drawn from the priors and from the closest empirical estimates — including the reading-speed–ability association ($r \approx .39$–$.48$) reported by Sortkær et al. (2021) and typical syntactic-complexity effects on children's reading ($d \approx 0.2$–0.5) — and will fit the confirmatory models (below) to each. For every focal parameter we will report the expected posterior width and the probability that its

95% HDI will fall determinately inside or outside the ROPE, that is, the probability of a conclusive practical-equivalence decision. This establishes a priori that, at the anticipated sample size, the design can return an informative answer to each hypothesis. The same simulation identifies the minimum analysable N at which a conclusive decision is reached with at least 90% probability for each parameter; if extraction yields fewer eligible observations for a given hypothesis — for example, within smaller language groups — that hypothesis will be flagged as insufficiently sensitive and interpreted descriptively rather than confirmatorily.

2.6.3 Outcome-Neutral Quality Checks and Positive Controls

To confirm that the data and the analysis pipeline can detect known effects before the focal hypotheses are evaluated, we pre-specify the following outcome-neutral checks, each independent of the focal hypotheses. First, as a validity check on the timing measure, total story reading time must increase with story word count; a strong positive association is a precondition for treating reading time as an index of reading effort, and its absence would indicate that recorded times do not reflect engagement with the text. Second, as a positive control on the feature pipeline, word frequency is the most robust and cross-linguistically replicable lexical predictor of reading time (Kuperman et al., 2024; van Heuven et al., 2014); the frequency-bearing component must show the expected slowing of reading for lower-frequency words, and its failure would indicate that the multilingual feature-extraction pipeline is not capturing genuine lexical signal, rendering null results for the other text components uninterpretable. Third, the per-language distribution of reading time per word will be screened for floor and ceiling effects and for adequate variance; an outcome lying at floor or ceiling for more than 90% of observations within a language will not be modelled in that language. These checks are evaluated before the confirmatory models; where a check fails, the dependent confirmatory test is reported as uninterpretable rather than as evidence for or against the corresponding hypothesis.

2.6.4 Dimension Reduction via Principal Component Analysis

Before fitting any mixed-effects model, we will reduce the full set of text-level features to a smaller number of stable, interpretable components using PCA, following the approach of Crossley (2025). PCA is chosen over exploratory factor analysis because the goal is dimension reduction of observed stimulus properties rather than estimation of unobserved latent causes: there is no theoretical basis for claiming that a latent factor *causes* MTLD, dependency distance and connective density to co-vary, and treating them as reflective indicators of a latent construct would misrepresent the nature of computational linguistic indices (Crossley, 2025).

Prior to PCA, variables will pass through several pre-screening stages. Variables with more than 20% missing values in any language, near-zero variance (SD < 0.01 after within-language standardisation), or excessive zero inflation (more than 20% zero counts) will be excluded. Variables showing absolute skewness or excess kurtosis greater than 3 will also be removed (Crossley, 2025). Where two variables correlate at $r > .90$, we will retain the variable with stronger theoretical motivation, better cross-linguistic availability, lower missingness, or clearer interpretability. This selection will not consult the variable's correlation with the outcome, preventing outcome information from leaking into feature construction.

All retained variables will be z-scored within language before PCA, removing between-language scale differences while preserving within-language variation. The primary analysis pools all language-standardised variables into a single PCA conducted in R using the `psych` package (Revelle, 2023), after verifying the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy (individual KMO > .50; overall KMO > .60) and Bartlett's test of sphericity. We use oblique rotation (promax,

$\kappa = 4$) rather than varimax, because linguistic features are naturally correlated and an orthogonality assumption is theoretically unjustified (Biber, 1988; Crossley, 2025). Components will be retained based on three jointly applied criteria: parallel analysis (Horn, 1965), which retains components whose eigenvalues exceed the 95th percentile from 10,000 matched random data structures; scree-plot inspection; and interpretability, with a component accepted only if the pattern matrix admits a coherent reading-theoretic interpretation. Variables with absolute pattern loadings below .30 will not be attributed to any component. Weighted regression factor scores for each retained component will then be computed per story and will serve as text-level predictors in the confirmatory models below. The full pattern matrix (all loadings ≥ .15), component-score intercorrelations, proportion of variance per component, and per-variable KMO values will be reported in a supplementary table, with each component named based on its highest-loading variables and theoretical interpretation.

2.6.5 Confirmatory Models

The confirmatory model predicts log-transformed reading time per word — log transformation normalises the positively skewed distribution, with posterior predictive fit verified via `pp_check` plots — from the retained text-feature PC scores and within-child reading experience (lagged log cumulative words read, person-mean-centred). The random-effects structure follows a maximal justified specification, with random intercepts for Child and Language and random slopes for the key PC components and for reading experience within Child (e.g., `(1 + SyntaxPC + CoherencePC + Experience | Child)`). Observations with standardised residuals exceeding ±3 *SD* based on posterior predictive draws will be flagged and removed, with both the full-data and trimmed models reported.

2.6.6 Convergence and Multicollinearity

A random intercept for *Language* will be included in all models to account for between-language variability and enable partial pooling. If the maximal random-effects structure fails to converge — assessed via $\widehat{R}$ and divergent transition diagnostics — we will simplify by removing the random slope with the smallest posterior variance and iterating until convergence is achieved, reporting all simplification steps. Residual multicollinearity among the retained PC scores will be assessed using variance inflation factors (VIF) on the design matrix after the dimension-reduction step; if any VIF exceeds 5, the component with the lowest theoretical relevance for the confirmatory hypotheses will be removed.

2.6.7 Model Comparison

We will compare the full model against reduced models that drop the text-feature components and the reading-experience term in turn, using leave-one-out cross-validation (LOO-CV; `loo` package), reporting the expected log pointwise predictive density difference ($\mathrm{ELPD}_{\mathrm{LOO}}$) and its standard error. LOO-CV is preferred over likelihood-ratio tests because it requires no asymptotic approximations and is well-calibrated within the Bayesian framework adopted here.

2.6.8 Minimum Effect Size and ROPE Calibration

ROPE bounds are informed by standardised effect sizes from the closest available prior studies: Sortkær et al. (2021) reported $r \approx .39–.48$ for reading-speed effects, and syntactic complexity effects on children's reading times in eye-tracking studies are typically in the range d = 0.2–0.5. A standardised β of 0.05 is therefore conservatively below the range of effects the existing literature considers educationally meaningful. Sensitivity analyses varying the ROPE bound to ±0.03 and ±0.10 will assess the robustness of practical-equivalence conclusions.

2.6.9 Sensitivity Analyses

Seven pre-registered sensitivity analyses will assess the robustness of the main findings. The primary models will first be re-fitted without the outlier exclusion criteria to establish whether winsorisation and residual trimming drive any substantive conclusions. They will then be restricted to the five largest language groups to evaluate cross-linguistic generalisability independently of less-represented languages. A third will vary the ROPE bounds to ±0.03 and ±0.10 as described above. In a fourth analysis, the confirmatory models will be re-estimated using sparse PCA scores — obtained via the R package `sparsepca`, with sparsity regularisation selected by cross-validation — to assess whether structurally simpler components with fewer non-zero loadings alter the substantive pattern of results. Language-specific PCAs will replace the pooled solution, with one PCA fitted per language, to test whether the cross-lingual feature space adequately represents within-language variation. A further analysis will restrict the sample to children who completed a higher minimum number of sessions (a completer-only subsample), to assess whether selective retention biases the within-child reading-experience effect (H2). Finally, a prior-sensitivity analysis will re-fit the confirmatory models under the previously specified weakly informative priors (Student's $t(3, 0, 2.5)$ on the fixed effects) to confirm that the substantive conclusions are driven by the data rather than by the informative priors (van de Schoot et al., 2021).

2.6.10 Supplementary Frequentist Models

To facilitate comparison with prior literature and to assist readers unfamiliar with Bayesian inference, fixed-effect point estimates, standard errors, and confidence intervals from a maximum-likelihood linear mixed model (LMM; fitted with `lme4`/`lmerTest`, identical random-effects structure) will be reported in a supplementary table. These results are descriptive only and carry no confirmatory weight; all hypothesis evaluation is conducted solely via the Bayesian framework described above.

3 Hypotheses

Our hypotheses are directional and each maps onto a specific fixed-effect term in the confirmatory model described above. Following the PCA procedure described in the Analytic Plan (see Dimension Reduction via Principal Component Analysis), hypotheses about text-level features are stated in terms of the PC component families expected to carry the relevant variance. The syntactic-structural component is referred to as SyntaxPC; the semantic/coherence component as CoherencePC; and the lexical-distributional component as LexDistPC. Because exact component names will be assigned post-hoc from the pattern matrix, each directional prediction is conditional on a component with the expected feature profile emerging from the PCA; if a relevant component does not emerge, the corresponding hypothesis will be reported as untestable.

First, we predict that text-level features predict reading time: stories scoring higher on syntactic and lexical difficulty — with longer dependency distances, less frequent words, higher lexical diversity and lower coherence — should be associated with longer reading times per word. We expect main effects of SyntaxPC and LexDistPC in the direction of longer reading times, and a main effect of CoherencePC in the direction of shorter reading times.

Second, we predict that children read progressively faster as they accumulate reading experience within the app. Reading practice builds fluency and lowers reading time over time, as indicated by app-based reading logs and longitudinal studies of reading experience (Schmidtke et al., 2024; Sortkær et al., 2021), so reading time per word should decrease as a child reads more. We operationalise within-child experience primarily as the cumulative number of words the child has read in prior sessions, lagged so the current story is excluded, log-transformed to capture the diminishing-returns shape of practice, and centred within child (person-mean), so that the effect is estimated purely within child and is not confounded with between-child differences in total app usage (Kraemer et al., 2000); the simpler session-order count (sequence position) is retained as a robustness coding. We use cumulative words rather than cumulative reading time, because the latter is constructed from the dependent variable. We expect a negative main effect of accumulated reading on reading time, with a by-child random slope capturing individual differences in practice rate. Because retention in the app is self-selected, this effect is associational — estimated within the children who remain and net of the text-feature components already in the model — and is accompanied by a completer-only sensitivity analysis (see Sensitivity Analyses).

Each hypothesis will be evaluated through the corresponding mixed-model term using the Bayesian inference criteria described in the Analytic Plan (posterior mean, 95% HDI, directional posterior probability, and ROPE analysis). Both hypotheses are evaluated under the same inferential framework, applied unconditionally and simultaneously.

3.1 Study Design Table

Table 4

Study design table linking research questions, hypotheses, statistical tests, and the interpretation of possible outcomes. All confirmatory tests use the Bayesian multilevel models and the ROPE/HDI decision rule set out in the Analytic Plan; sampling adequacy for each test is established by the design analysis (see Sampling Plan and Inferential Sensitivity), and every test is conditional on the outcome-neutral quality checks.

Research question	Hypothesis (predicted effect)	Test (model term and inference)	Interpretation of possible results
RQ1 — text properties predict reading time per word	**H1.** Higher syntactic and lexical difficulty lengthen, and higher coherence shortens, reading time: β(SyntaxPC) > 0, β(LexDistPC) > 0, β(CoherencePC) < 0.	Main effects in the primary reading-time model (log reading time per word); posterior mean, 95% HDI, directional probability, and ROPE. Conditional on the word-count and frequency positive controls.	For each component, a 95% HDI excluding the ROPE in the predicted direction supports H1; an HDI inside the ROPE indicates no practically meaningful effect; an HDI straddling a ROPE bound is inconclusive.
RQ2 — reading changes with accumulated experience	**H2.** Reading speeds up as within-child experience accumulates: β(Experience) < 0 on reading time.	Main effect of within-child reading experience (lagged log cumulative words read, person-mean-centred) in the reading-time model, with a by-child random slope; posterior mean, 95% HDI, directional probability, and ROPE. Associational, with a completer-only sensitivity analysis.	An HDI below the ROPE in the negative direction supports H2; an HDI inside the ROPE indicates no practically meaningful experience effect; a straddling HDI is inconclusive.

3.2 Exploratory Analyses

Beyond the confirmatory tests above, we plan several analyses that are explicitly designated as exploratory and will be reported separately.

We will examine whether the magnitude of text-feature effects on children's reading varies systematically across the 19 languages represented in the *Let's Story* dataset. This could be addressed through language-stratified models or through meta-regression with typological features (e.g., orthographic depth) as moderators — an approach that may reveal whether the relationships between, say, syntactic complexity and reading time are consistent across typologically diverse languages or are modulated by language-specific properties.

We will also explore whether text complexity features show non-linear (quadratic) relationships with reading time, which would suggest diminishing or accelerating effects at extreme levels of complexity.

Two further exploratory analyses build on the confirmatory within-child experience effect (H2). First, although the reviewed evidence that reading experience attenuates sensitivity to text difficulty concerns between-child skill differences rather than within-child change over a short window, we will explore whether accumulated experience moderates the text-feature effects — most plausibly attenuating the lexical-distributional cost, in line with the smaller lexical and word-length effects shown by more fluent and experienced readers (Hsiao & Nation, 2018; Lubineau et al., 2024) — while remaining agnostic about the syntactic and coherence components, for which the reviewed corpus is silent. Second, because practice curves are typically non-linear, we will compare the primary log-cumulative-words parameterisation against linear and by-child smooth (generalised additive) alternatives, and against the simpler session-order coding (Baayen et al., 2021). Throughout, accumulated reading is treated cautiously: it is confounded with selective retention and with drift in the difficulty of the stories served, so we verify that the confirmatory text-feature effects are unchanged when the experience term and its by-child slope are included, alongside the completer-only check reported under Sensitivity Analyses.

All exploratory results will be clearly labelled as such in the manuscript, and we will not draw confirmatory conclusions from these analyses.

Declarations

Ethics

This study is a secondary analysis of anonymised behavioural data generated during ordinary use of a commercial application; parental or guardian consent for data collection was registered in the app at account creation, under the platform's terms of use and privacy policy. No data are collected from children specifically for this study beyond those produced by normal app use, and the research team accesses no personally identifying information. Ethical review for the secondary analysis [has been / will be] obtained from the Departmental Research Ethics Committee of the University of Oxford's Department of Education, operating under the Central University Research Ethics Committee (CUREC) framework [approval reference to be inserted]. Processing of these data, which concern minors, complies with the applicable data-protection law (including the UK General Data Protection Regulation); the data controller for the source data is [Ferrero International and/or Gameloft — to be confirmed].

Data and Code Availability

In keeping with the Registered Reports format, this Stage 1 protocol will be deposited in a public repository and made available either immediately or upon Stage 2 acceptance. All analysis code, the story-level computational text-feature matrices, and the aggregated data required to reproduce the reported results will be made openly available (for example, on the Open Science Framework) at Stage 2. The raw, individual-level reading logs are the property of the data controller ([Ferrero International and/or Gameloft]) and contain data from minors; their release is therefore governed by the data-sharing agreement and by data-protection law, and they will be made available from the data controller on reasonable request to the extent those constraints permit [author to confirm what may be shared]. This statement complements the outcome-blind data-access certification given under Data Provenance and Researcher Access.

Funding

This work is supported by Kinder and the John Fell Fund (University of Oxford) [grant numbers to be inserted]. The behavioural data analysed here were generated by the *Applaydu* platform in the course of its normal operation, independently of this study and prior to its conception. Beyond providing access to these pre-existing, anonymised data, the funders and the data provider had no role in the design of the study, in the analysis or interpretation of the data, or in the decision to submit the work for publication.

Competing Interests

The behavioural data analysed here are generated by *Applaydu*, an application developed by Gameloft and Ferrero International, and the research is funded in part by Kinder, a brand owned by Ferrero. This dual relationship — the funder and the provider of the data being the same commercial group whose product is under study — constitutes a potential competing interest, which the authors declare. Any further interests (for example, employment, consultancy, or other financial relationships with Ferrero or Gameloft) must also be declared; if none exist beyond the funding and data-provision relationship described here, this should be stated explicitly [author to confirm].

Appendix: Per-Language Norm Sources

For the lexicon-dependent measures (frequency, age of acquisition, concreteness/imageability, valence/arousal, and the LIWC word-category counts), the validated native norms listed in Table 5 will be used wherever they exist. Every other language–measure combination defaults to the cross-language resource specified in Table 3 — `wordfreq` for frequency (Speer, 2022), the NRC VAD lexicon for valence and arousal (Mohammad, 2018), and the 25-language set of Łuniewska et al. (2016) for age of acquisition — or, where no validated norm is available, is recorded as missing and removed by the pre-screening rule (see Analytic Plan). Native frequency norms are additionally used to validate the `wordfreq` estimates. The language-general measures (lexical diversity, all syntactic indices, coherence, and surprisal) are computed identically across languages and are not repeated here.

Table 5

Validated native norm sources by language for the lexicon-dependent measures. Entries not listed use the cross-language defaults described in the text (`wordfreq` for frequency, the NRC VAD lexicon for valence and arousal, and the Łuniewska et al. (2016) set for age of acquisition) or are treated as missing under the pre-screening rule.

Language	Validated native norms (cross-language defaults used otherwise)
English	Frequency, SUBTLEX (van Heuven et al., 2014); age of acquisition (Kuperman et al., 2012); concreteness (Brysbaert, Warriner, et al., 2014); valence and arousal (Warriner et al., 2013); LIWC-22 dictionary (Boyd et al., 2022).
Spanish	Frequency, EsPal (Duchon et al., 2013); age of acquisition (Alonso et al., 2015); valence and arousal (Stadthagen-González et al., 2017); LIWC Spanish dictionary.
German	Frequency, SUBTLEX-DE (Brysbaert et al., 2011); valence, arousal and imageability from BAWL-R (Võ et al., 2009) and the automatically generated norm set (Köper & Schulte im Walde, 2016); LIWC German dictionary.
Dutch	Frequency, SUBTLEX-NL (Keuleers et al., 2010); age of acquisition and concreteness (Brysbaert, Stevens, et al., 2014); valence and arousal (Moors et al., 2013); LIWC Dutch dictionary.

Language	Validated native norms (cross-language defaults used otherwise)
French	Frequency, Lexique (New et al., 2001); age of acquisition (Ferrand et al., 2008); imageability (Desrochers & Thompson, 2009); valence and arousal from FAN (Monnier & Syssau, 2014) and, for children and adolescents, FANchild (Monnier & Syssau, 2017); LIWC French dictionary.
Italian	Frequency, SUBTLEX-IT (Crepaldi et al., 2015); concreteness, imageability and age of acquisition (Della Rosa et al., 2010); valence, arousal and dominance (Montefinese et al., 2014); LIWC Italian dictionary.
Portuguese	Frequency, SUBTLEX-PT-BR (Tang, 2012); imageability, concreteness and subjective frequency from the Minho Word Pool (Soares et al., 2017); valence, arousal and dominance (Soares et al., 2012); LIWC (Brazilian) Portuguese dictionary.
Polish	Frequency, SUBTLEX-PL (Mandera et al., 2015); valence, arousal, concreteness, imageability and age of acquisition from ANPW_R (Imbir, 2016).
Chinese (Mandarin)	Frequency, SUBTLEX-CH (Cai & Brysbaert, 2010); valence and arousal from CVAW (Yu et al., 2016); LIWC Chinese dictionary.
Russian, Turkish, Romanian	Frequency from `wordfreq`; age of acquisition from the cross-language set of Łuniewska et al. (2016) (these languages are among its 25); valence and arousal from NRC VAD; a published LIWC translation dictionary; concreteness and imageability treated as missing.
Hebrew (and Greek, if present in the corpus)	Frequency from `wordfreq`; age of acquisition from the cross-language set of Łuniewska et al. (2016); valence and arousal from NRC VAD; concreteness, imageability and LIWC categories treated as missing.
Korean, Indonesian, Thai, Arabic	Frequency from `wordfreq`; valence and arousal from NRC VAD; age of acquisition and concreteness generally unavailable and treated as missing. For Thai, and any language without reliable automatic word segmentation, the dependency-based syntactic features are verified before use and that language is excluded from the affected analyses if they prove inadequate (see Materials).

The exact set of generation languages, and hence the rows of this appendix, will be fixed from the platform metadata at data extraction (see Participants).

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